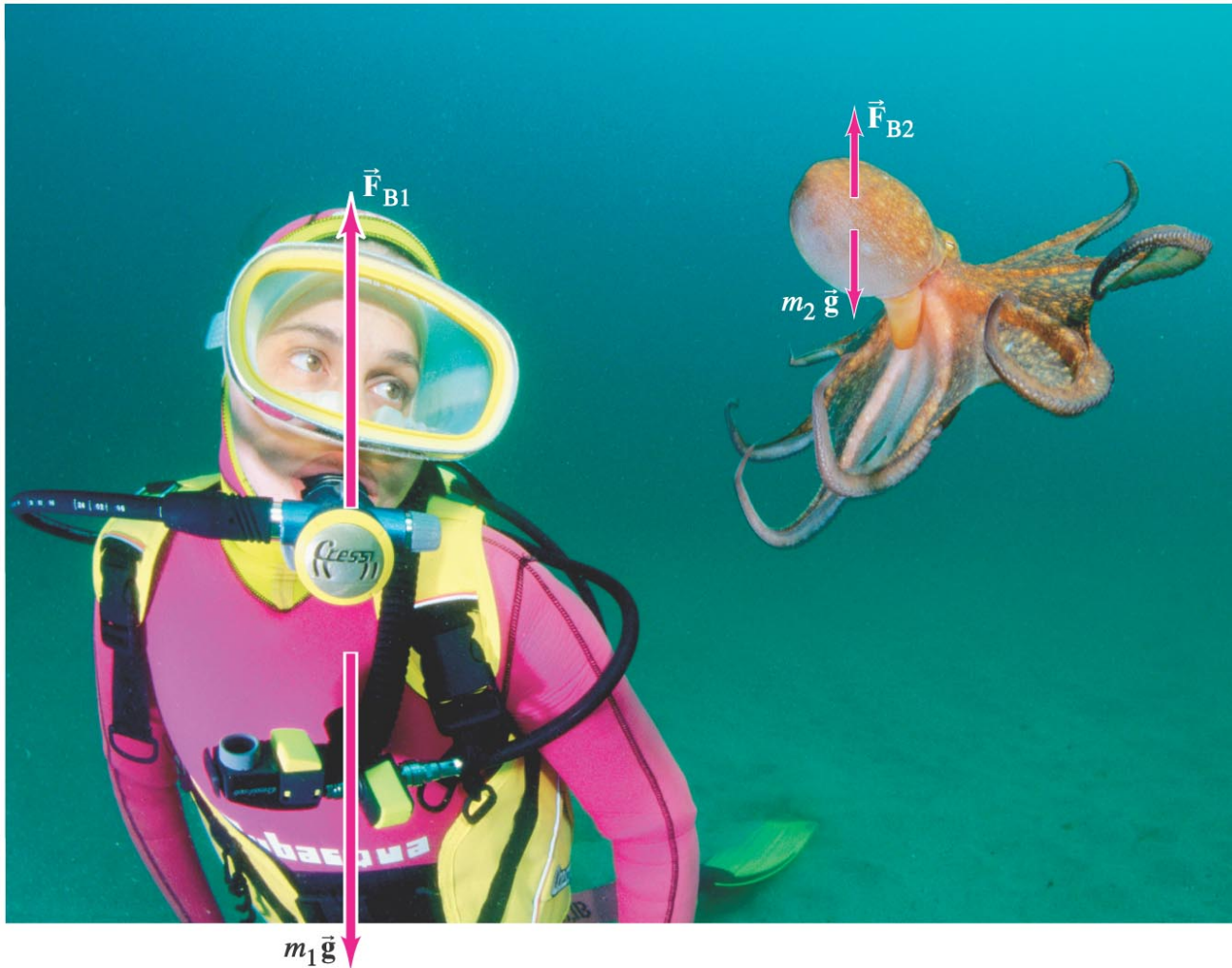


Chapter 13

Fluids



Units of Chapter 13

- **Density and Specific Gravity**
- **Pressure in Fluids**
- **Atmospheric Pressure and Gauge Pressure**
- **Pascal's Principle**
- **Measurement of Pressure; Gauges and the Barometer**
- **Buoyancy and Archimedes' Principle**

Units of Chapter 13

- **Fluids in Motion; Flow Rate and the Equation of Continuity**
- **Bernoulli's Equation**
- **Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA**
- **Viscosity**
- **Flow in Tubes: Poiseuille's Equation, Blood Flow**
- **Surface Tension and Capillarity**

13-1 Phases of Matter

The three common phases of matter are **solid, liquid, and gas.**

A solid has a definite **shape and size.**

A liquid has a fixed **volume** but can be any shape.

A gas can be any shape and also can be easily **compressed.**

Liquids and gases both **flow**, and are called **fluids.**

13-2 Density and Specific Gravity

The density ρ of a substance is its mass per unit volume:

$$\rho = \frac{m}{V}.$$

The SI unit for density is kg/m³. Density is also sometimes given in g/cm³; to convert g/cm³ to kg/m³, multiply by 1000.

Water at 4°C has a density of 1 g/cm³ = 1000 kg/m³.

The **specific gravity** of a substance is the ratio of its density to that of water.

13-2 Density and Specific Gravity

Example 13-1: Mass, given volume and density.

What is the mass of a solid iron wrecking ball of radius 18 cm?

13-3 Pressure in Fluids

Pressure is defined as the force per unit area.

$$\text{pressure} = P = \frac{F}{A}.$$

Pressure is a scalar; the units of pressure in the SI system are pascals:

$$1 \text{ Pa} = 1 \text{ N/m}^2.$$

13-3 Pressure in Fluids

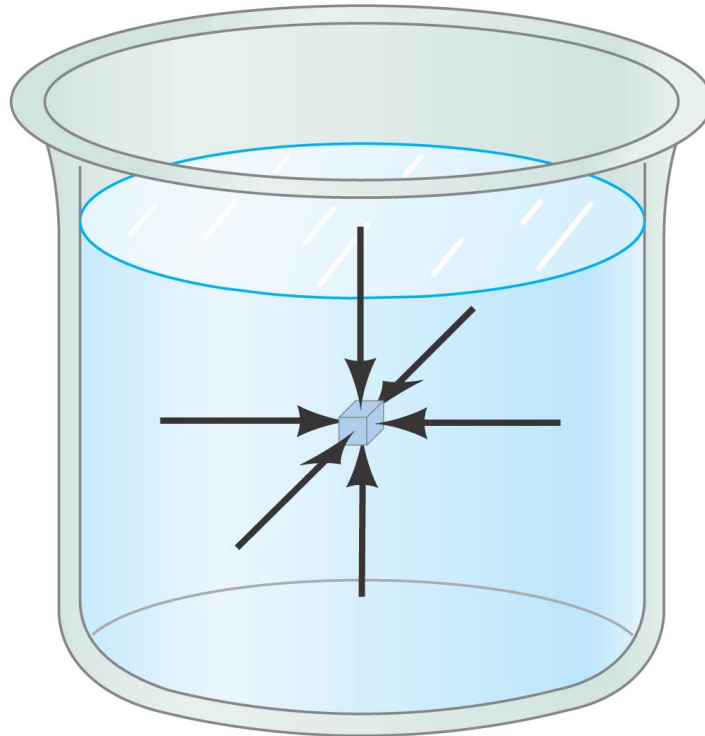
Example 13-2: Calculating pressure.

The two feet of a 60-kg person cover an area of 500 cm².

- (a) Determine the pressure exerted by the two feet on the ground.**
- (b) If the person stands on one foot, what will the pressure be under that foot?**

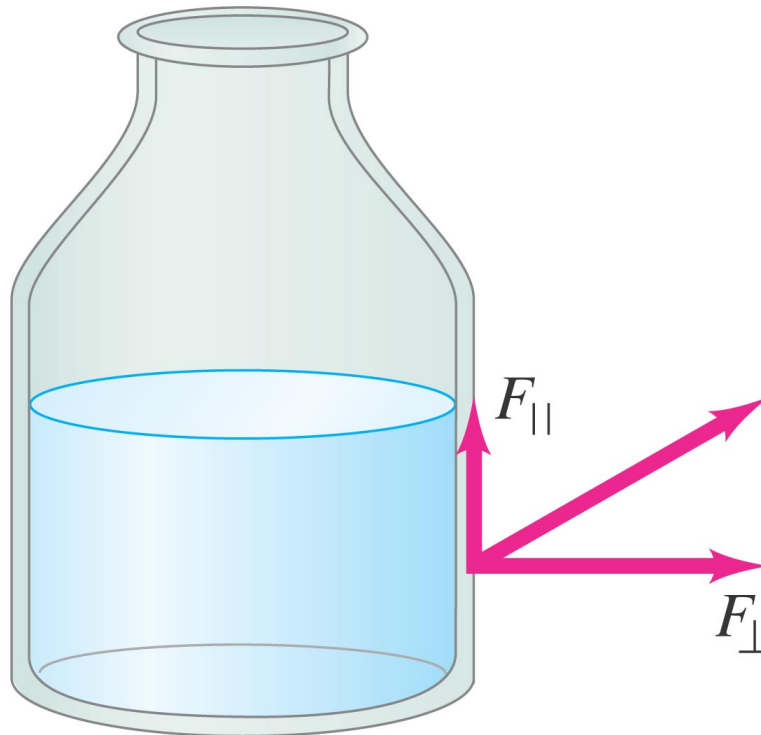
13-3 Pressure in Fluids

Pressure is the same in every direction in a static fluid at a given depth; if it were not, the fluid would flow.



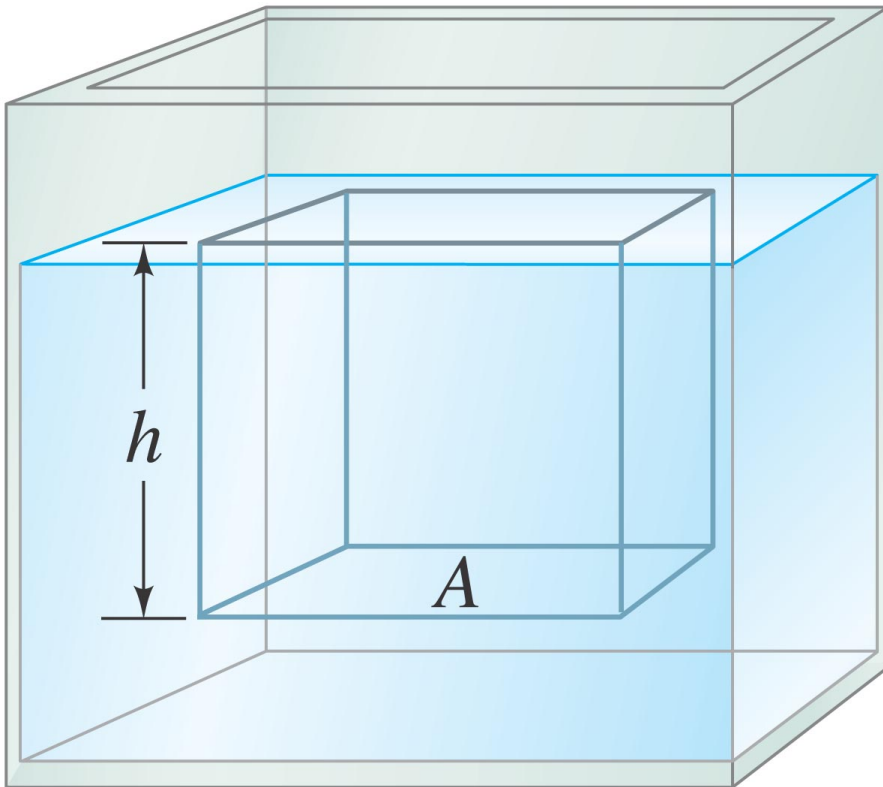
13-3 Pressure in Fluids

For a fluid at rest, there is also no component of force **parallel** to any solid surface—once again, if there were, the fluid would flow.



13-3 Pressure in Fluids

The **pressure** at a depth h below the surface of the liquid is due to the **weight** of the liquid above it. We can quickly calculate:



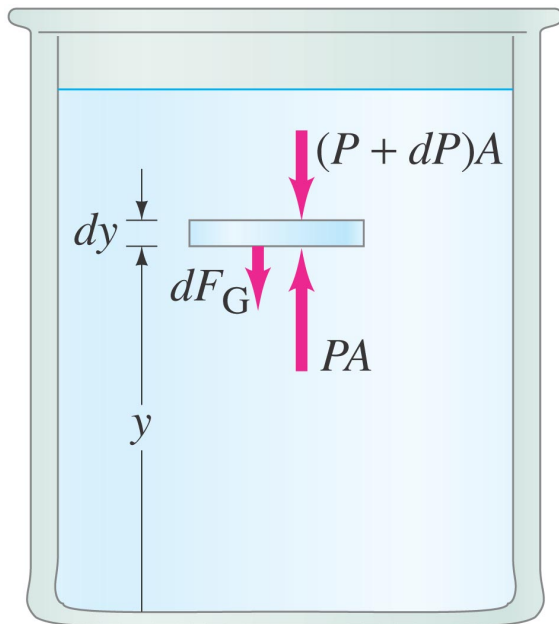
$$P = \frac{F}{A} = \frac{\rho Ahg}{A}$$

$$P = \rho gh.$$

This relation is valid for any liquid whose density does not change with depth.

13-3 Pressure in Fluids

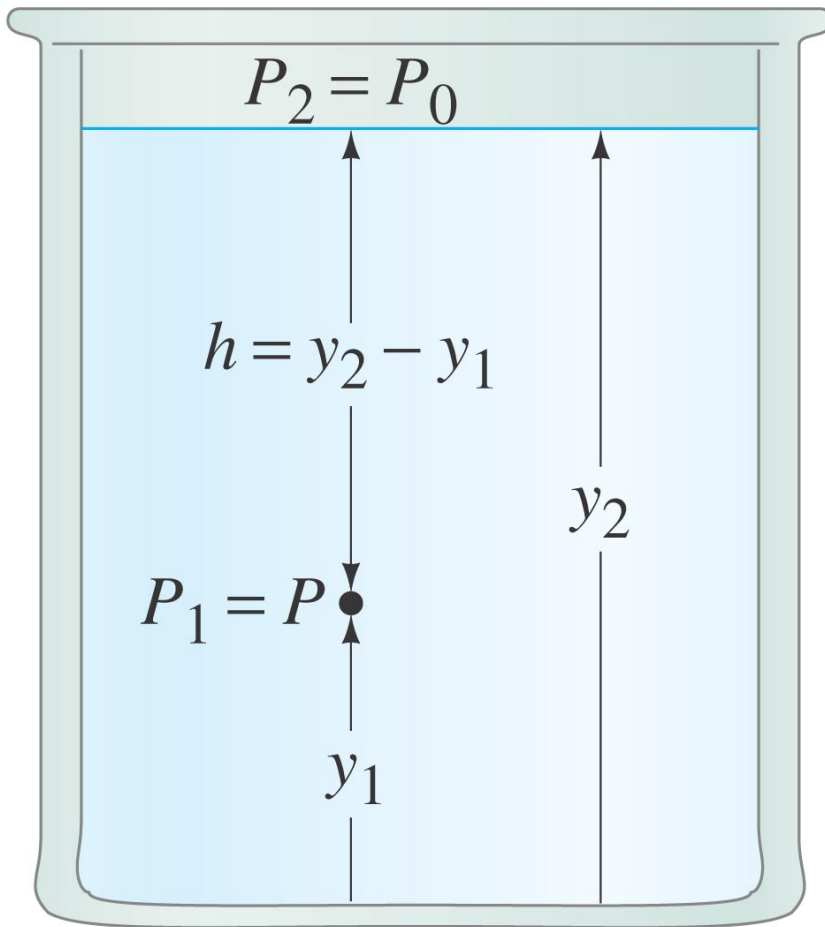
If there is external pressure in addition to the weight of the fluid itself, or if the density of the fluid is not constant, we calculate the pressure at a height y in the fluid; the negative sign indicates that the pressure decreases with height (increases with depth):



$$\frac{dP}{dy} = -\rho g.$$

13-3 Pressure in Fluids

We then integrate to find the pressure:

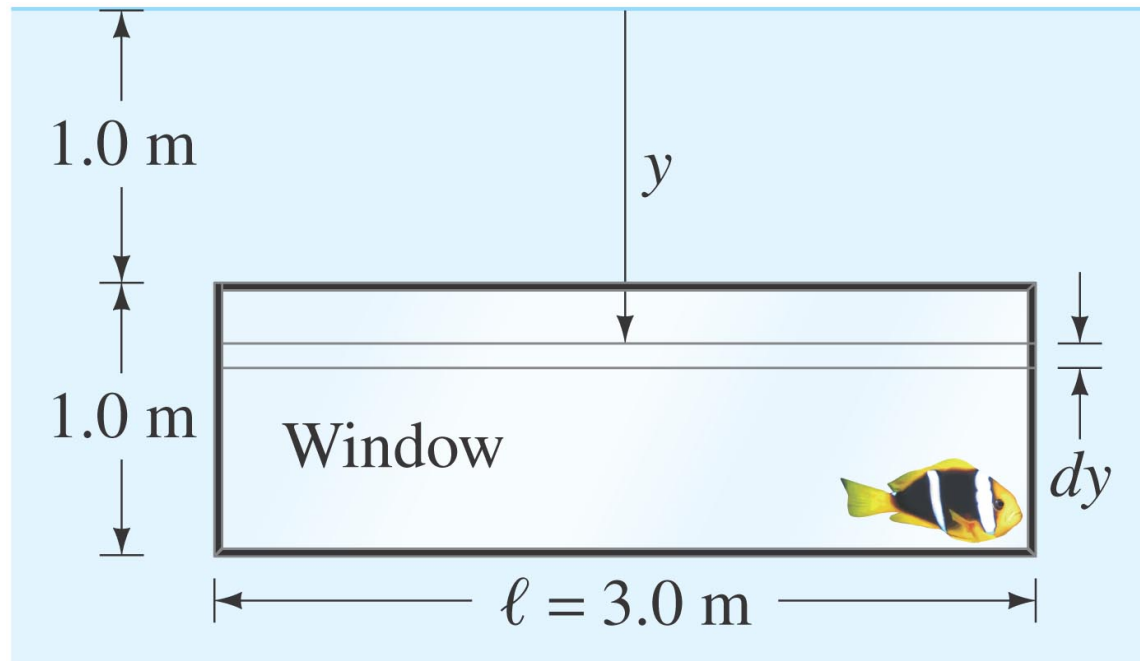


$$\int_{P_1}^{P_2} dP = - \int_{y_1}^{y_2} \rho g dy$$
$$P_2 - P_1 = - \int_{y_1}^{y_2} \rho g dy.$$

13-3 Pressure in Fluids

Example 13-4: Force on aquarium window.

Calculate the force due to water pressure exerted on a $1.0\text{ m} \times 3.0\text{ m}$ aquarium viewing window whose top edge is 1.0 m below the water surface.



13-3 Pressure in Fluids

Example 13-5: Elevation effect on atmospheric pressure.

- (a) Determine the variation in pressure in the Earth's atmosphere as a function of height y above sea level, assuming g is constant and that the density of the air is proportional to the pressure. (This last assumption is not terribly accurate, in part because temperature and other weather effects are important.)**
- (b) At what elevation is the air pressure equal to half the pressure at sea level?**

13-4 Atmospheric Pressure and Gauge Pressure

At sea level the atmospheric pressure is about $1.013 \times 10^5 \text{ N/m}^2$; this is called 1 atmosphere (atm).

Another unit of pressure is the bar:

$$1 \text{ bar} = 1.00 \times 10^5 \text{ N/m}^2.$$

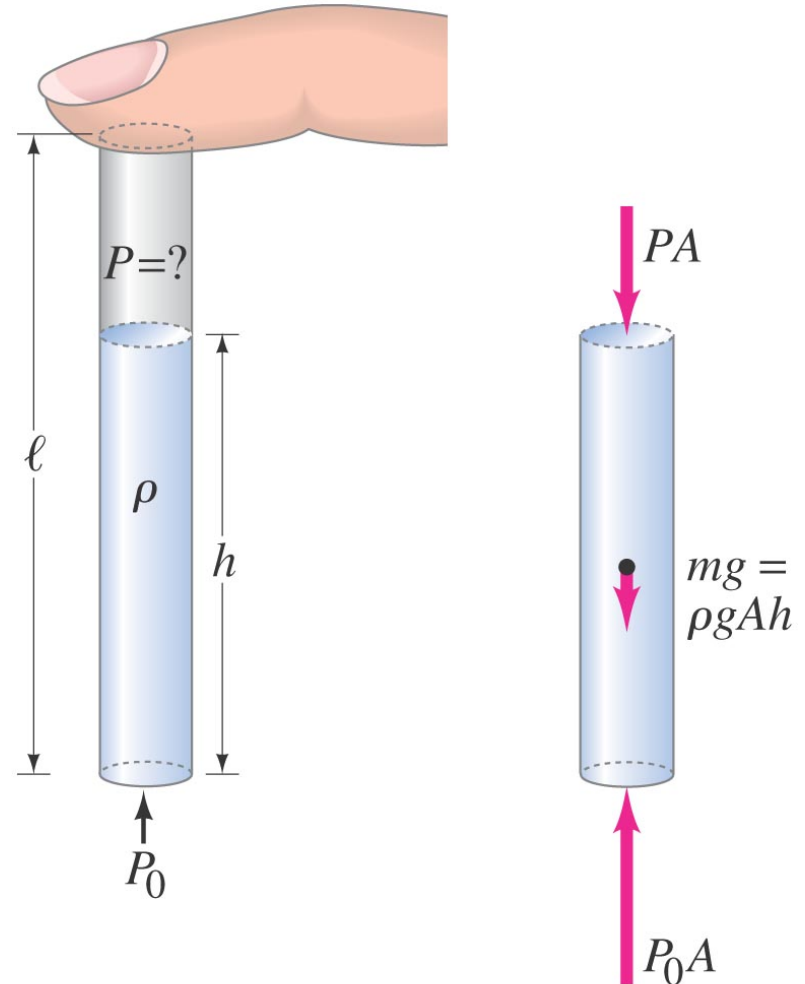
Standard atmospheric pressure is just over 1 bar.

This pressure does not crush us, as our cells maintain an internal pressure that balances it.

13-4 Atmospheric Pressure and Gauge Pressure

Conceptual Example 13-6: Finger holds water in a straw.

You insert a straw of length l into a tall glass of water. You place your finger over the top of the straw, capturing some air above the water but preventing any additional air from getting in or out, and then you lift the straw from the water. You find that the straw retains most of the water. Does the air in the space between your finger and the top of the water have a pressure P that is greater than, equal to, or less than the atmospheric pressure P_0 outside the straw?



13-4 Atmospheric Pressure and Gauge Pressure

Most pressure gauges measure the pressure above the atmospheric pressure—this is called the gauge pressure.

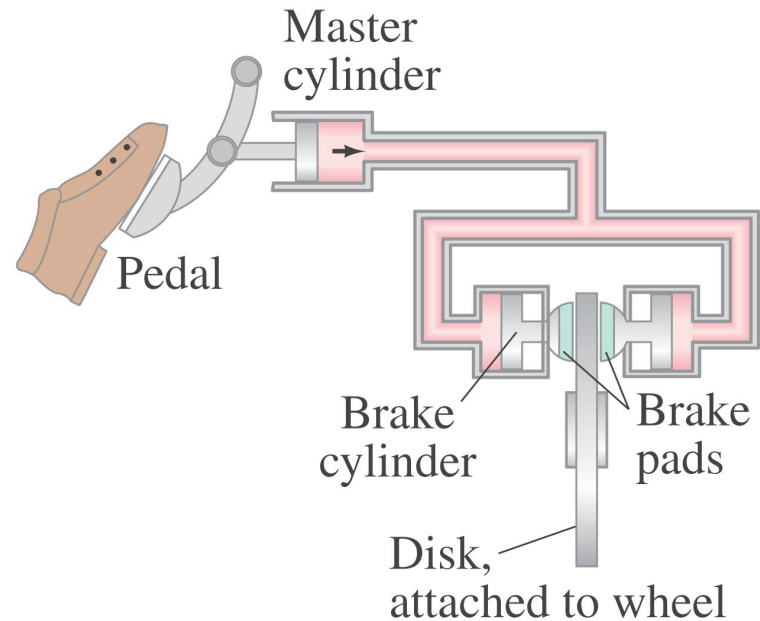
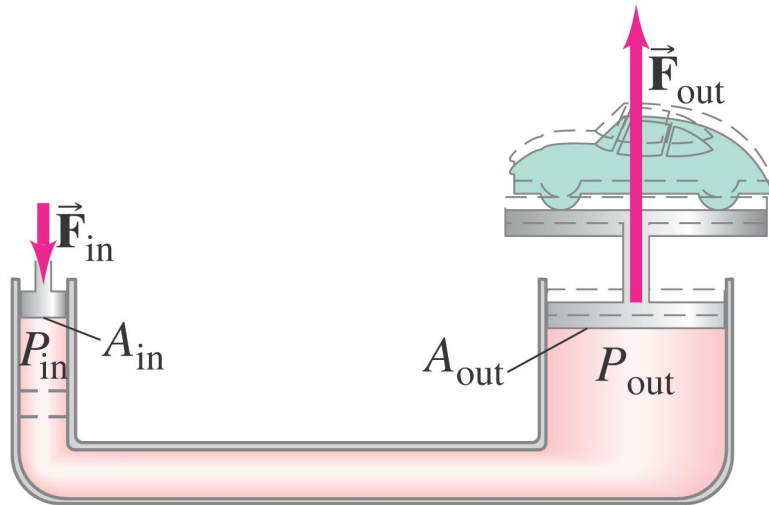
The absolute pressure is the sum of the atmospheric pressure and the gauge pressure.

$$P = P_0 + P_G.$$

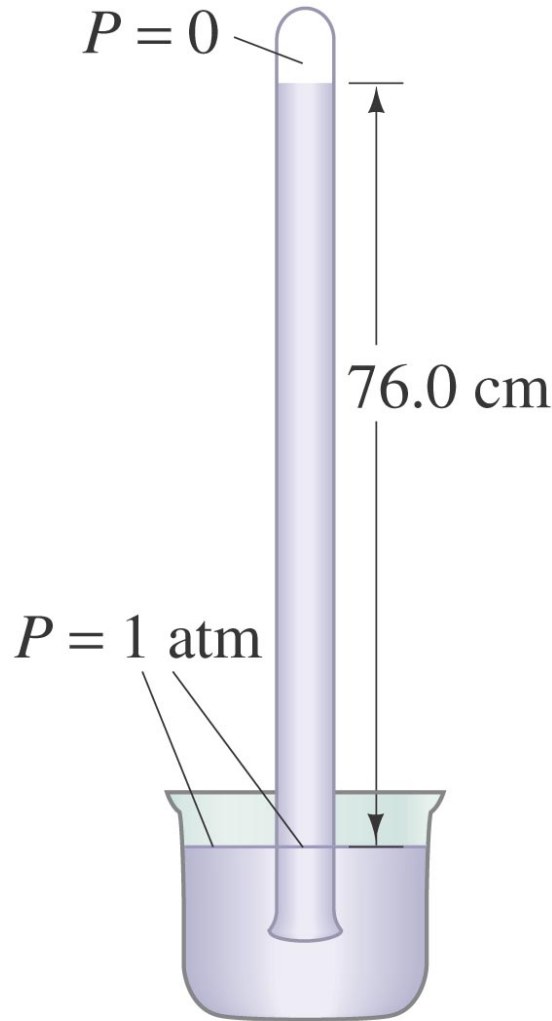
13-5 Pascal's Principle

If an **external pressure** is applied to a **confined fluid**, the pressure at **every point** within the fluid increases by that amount.

This principle is used, for example, in **hydraulic lifts** and **hydraulic brakes**.



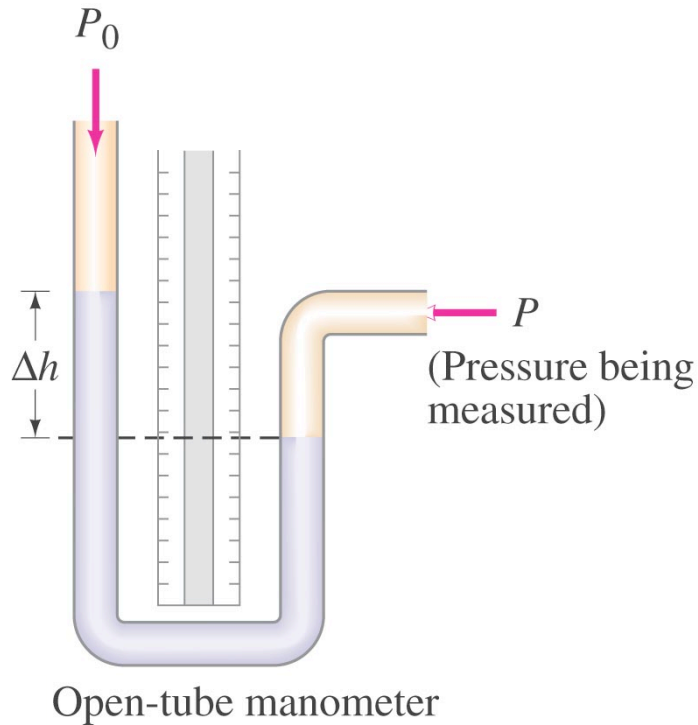
13-6 Measurement of Pressure; Gauges and the Barometer



This is a **mercury** barometer, developed by Torricelli to **measure atmospheric pressure**. The **height of the column of mercury** is such that the pressure in the tube at the surface level is **1 atm**.

Therefore, pressure is often quoted in **millimeters (or inches) of mercury**.

13-6 Measurement of Pressure; Gauges and the Barometer



There are a number of different types of **pressure gauges**. This one is an **open-tube manometer**. The **pressure in the open end is atmospheric pressure**; the **pressure being measured will cause the fluid to rise until the pressures on both sides at the same height are equal**.

13-6 Measurement of Pressure; Gauges and the Barometer



Any liquid can serve in a Torricelli-style barometer, but the most **dense** ones are the most convenient. This barometer uses **water**.

13-6 Measurement of Pressure; Gauges and the Barometer

Pressure is measured in a variety of different units. This table gives the conversion factors.

TABLE 13–2 Conversion Factors Between Different Units of Pressure

In Terms of $1 \text{ Pa} = 1 \text{ N/m}^2$	1 atm in Different Units
$1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$ $= 1.013 \times 10^5 \text{ Pa} = 101.3 \text{ kPa}$	$1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$
$1 \text{ bar} = 1.000 \times 10^5 \text{ N/m}^2$	$1 \text{ atm} = 1.013 \text{ bar}$
$1 \text{ dyne/cm}^2 = 0.1 \text{ N/m}^2$	$1 \text{ atm} = 1.013 \times 10^6 \text{ dyne/cm}^2$
$1 \text{ lb/in.}^2 = 6.90 \times 10^3 \text{ N/m}^2$	$1 \text{ atm} = 14.7 \text{ lb/in.}^2$
$1 \text{ lb/ft}^2 = 47.9 \text{ N/m}^2$	$1 \text{ atm} = 2.12 \times 10^3 \text{ lb/ft}^2$
$1 \text{ cm-Hg} = 1.33 \times 10^3 \text{ N/m}^2$	$1 \text{ atm} = 76.0 \text{ cm-Hg}$
$1 \text{ mm-Hg} = 133 \text{ N/m}^2$	$1 \text{ atm} = 760 \text{ mm-Hg}$
$1 \text{ torr} = 133 \text{ N/m}^2$	$1 \text{ atm} = 760 \text{ torr}$
$1 \text{ mm-H}_2\text{O} (4^\circ\text{C}) = 9.80 \text{ N/m}^2$	$1 \text{ atm} = 1.03 \times 10^4 \text{ mm-H}_2\text{O} (4^\circ\text{C})$

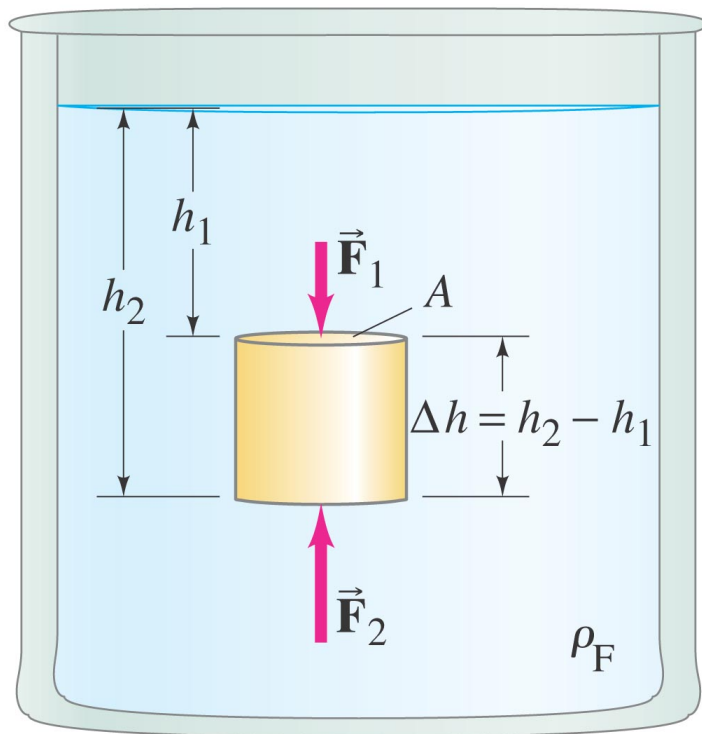
13-6 Measurement of Pressure; Gauges and the Barometer

Conceptual Example 13-7: Suction.

A student suggests suction-cup shoes for Space Shuttle astronauts working on the exterior of a spacecraft. Having just studied this Chapter, you gently remind him of the fallacy of this plan. What is it?

13-7 Buoyancy and Archimedes' Principle

This is an object **submerged** in a fluid. There is a **net force** on the object because the pressures at the top and bottom of it are different.



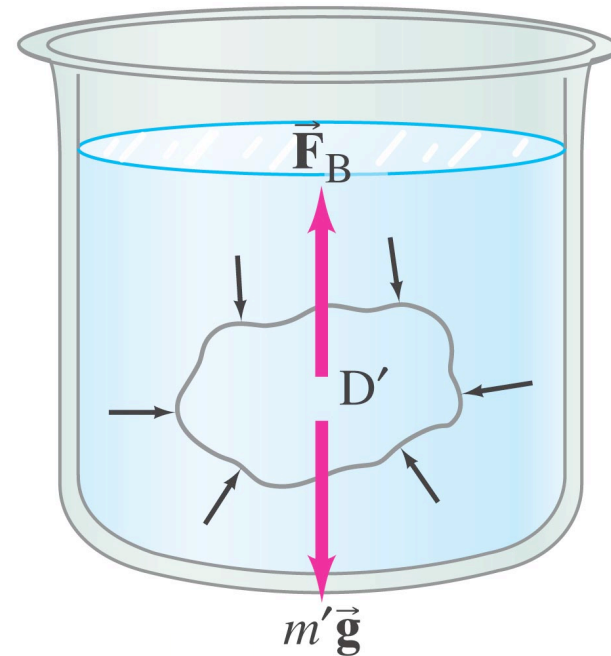
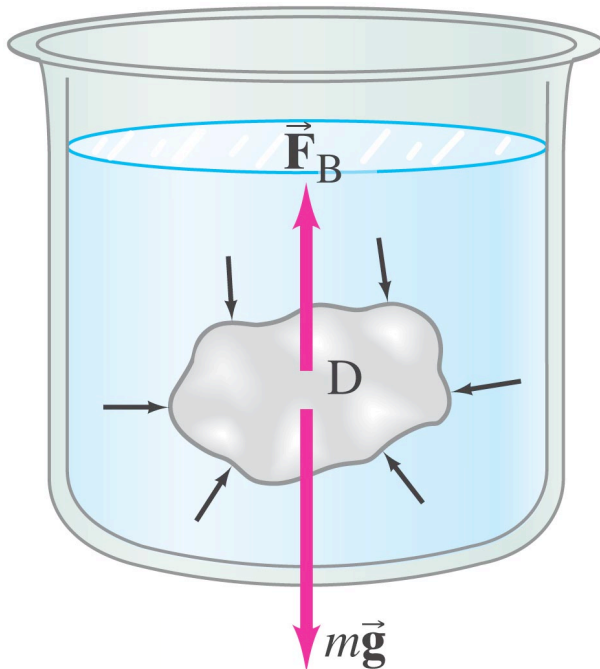
The buoyant force is found to be the upward force on the same volume of water:

$$\begin{aligned} F_B &= F_2 - F_1 = \rho_F g A (h_2 - h_1) \\ &= \rho_F g A \Delta h \\ &= \rho_F V g \\ &= m_F g. \end{aligned}$$

13-7 Buoyancy and Archimedes' Principle

Archimedes' principle:

The buoyant force on an object immersed in a fluid is equal to the weight of the fluid displaced by that object.



13-7 Buoyancy and Archimedes' Principle

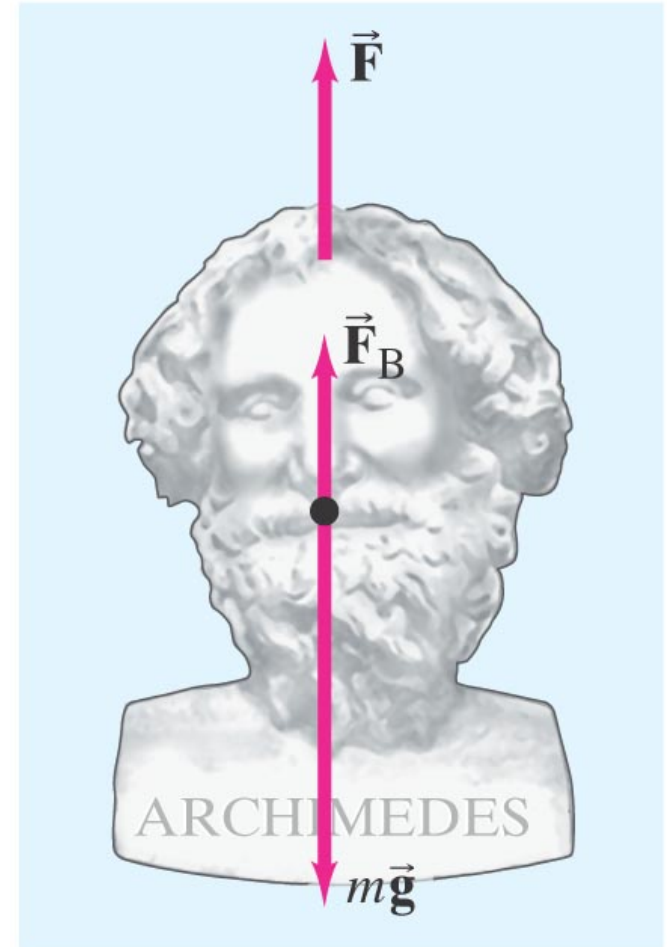
Conceptual Example 13-8: Two pails of water.

Consider two identical pails of water filled to the brim. One pail contains only water, the other has a piece of wood floating in it. Which pail has the greater weight?

13-7 Buoyancy and Archimedes' Principle

Example 13-9: Recovering a submerged statue.

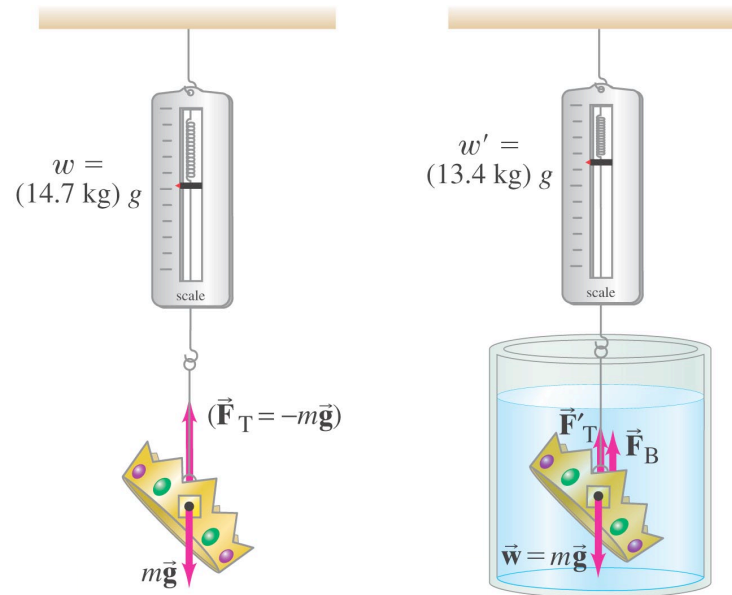
A 70-kg ancient statue lies at the bottom of the sea. Its volume is $3.0 \times 10^4 \text{ cm}^3$. How much force is needed to lift it?



13-7 Buoyancy and Archimedes' Principle

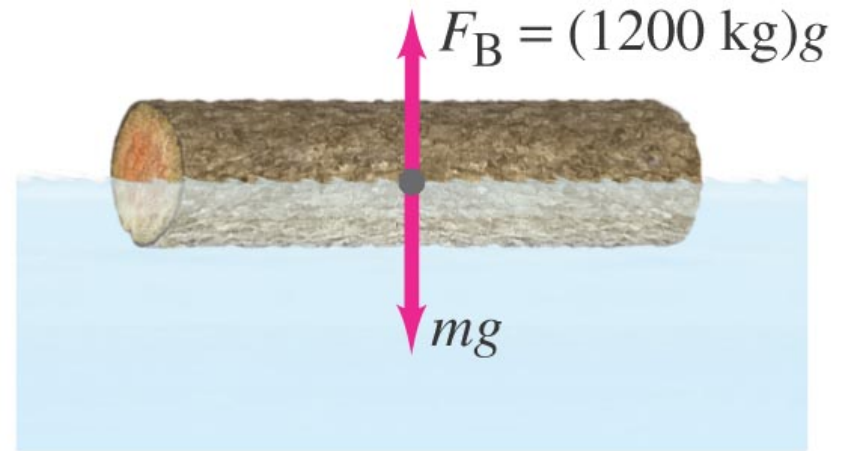
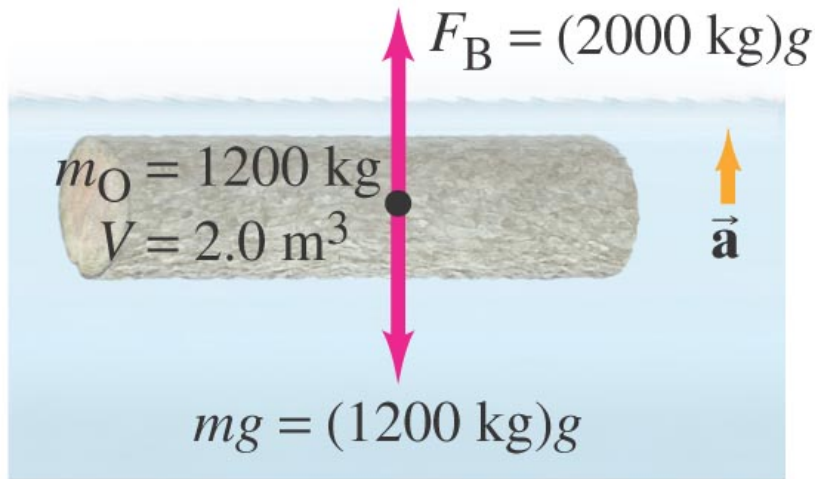
Example 13-10: Archimedes: Is the crown gold?

When a crown of mass 14.7 kg is submerged in water, an accurate scale reads only 13.4 kg. Is the crown made of gold?



13-7 Buoyancy and Archimedes' Principle

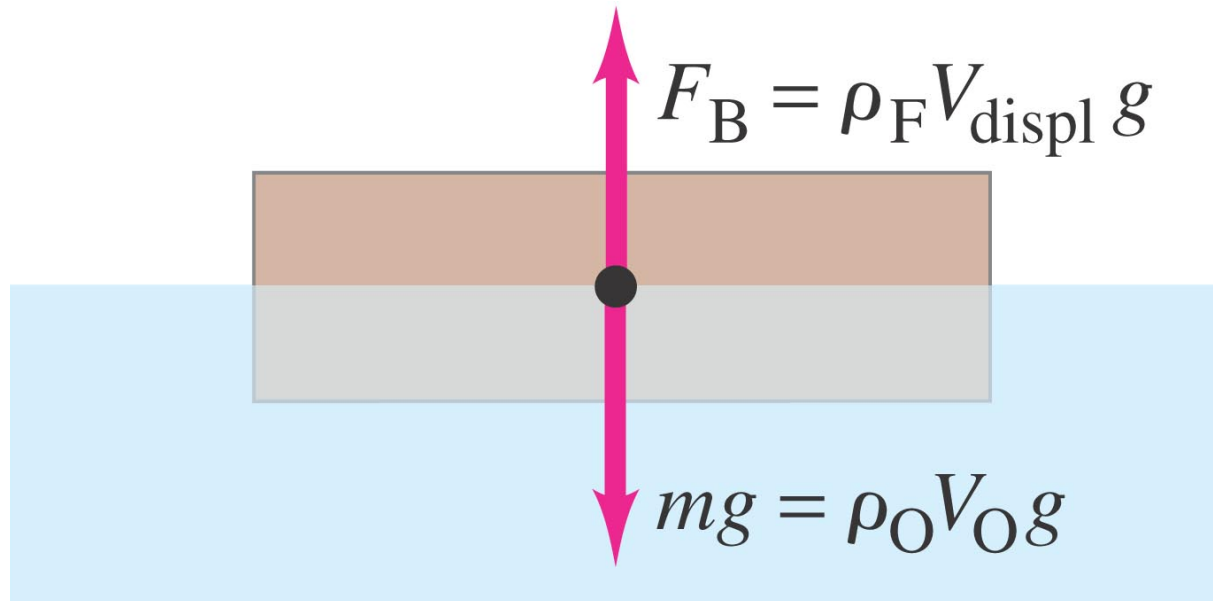
If an object's density is less than that of water, there will be an upward net force on it, and it will rise until it is partially out of the water.



13-7 Buoyancy and Archimedes' Principle

For a floating object, the fraction that is submerged is given by the ratio of the object's density to that of the fluid.

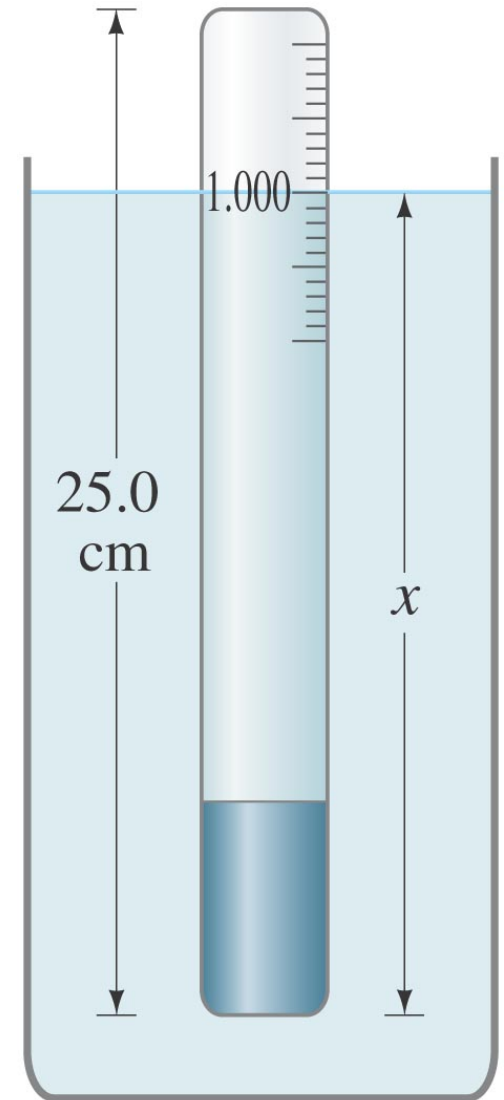
$$\frac{V_{\text{displ}}}{V_O} = \frac{\rho_O}{\rho_F}.$$



13-7 Buoyancy and Archimedes' Principle

Example 13-11: Hydrometer calibration.

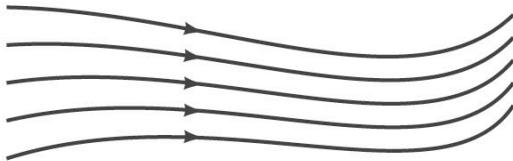
A hydrometer is a simple instrument used to measure the specific gravity of a liquid by indicating how deeply the instrument sinks in the liquid. This hydrometer consists of a glass tube, weighted at the bottom, which is 25.0 cm long and 2.00 cm^2 in cross-sectional area, and has a mass of 45.0 g. How far from the end should the 1.000 mark be placed?



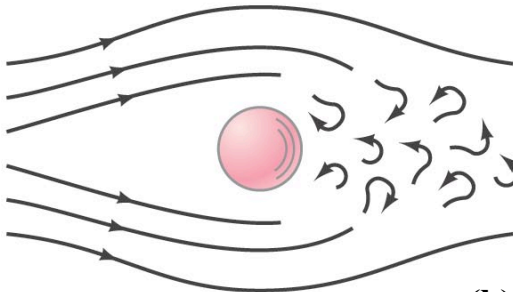
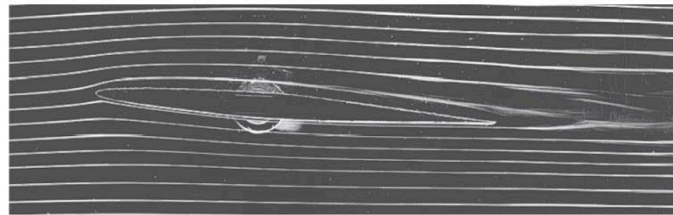
13-8 Fluids in Motion; Flow Rate and the Equation of Continuity

If the flow of a fluid is smooth, it is called **streamline or laminar flow (a)**.

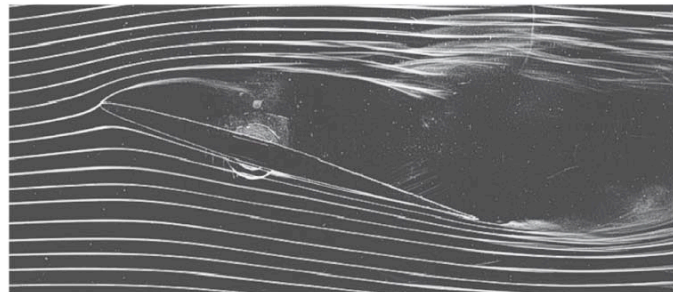
Above a certain speed, the flow becomes **turbulent (b)**. Turbulent flow has **eddies**; the **viscosity** of the fluid is much greater when eddies are present.



(a)



(b)



13-8 Fluids in Motion; Flow Rate and the Equation of Continuity

We will deal with **laminar flow**.

The **mass flow rate** is the mass that passes a given point per unit time. The flow rates at any two points must be **equal**, as long as no fluid is being added or taken away.

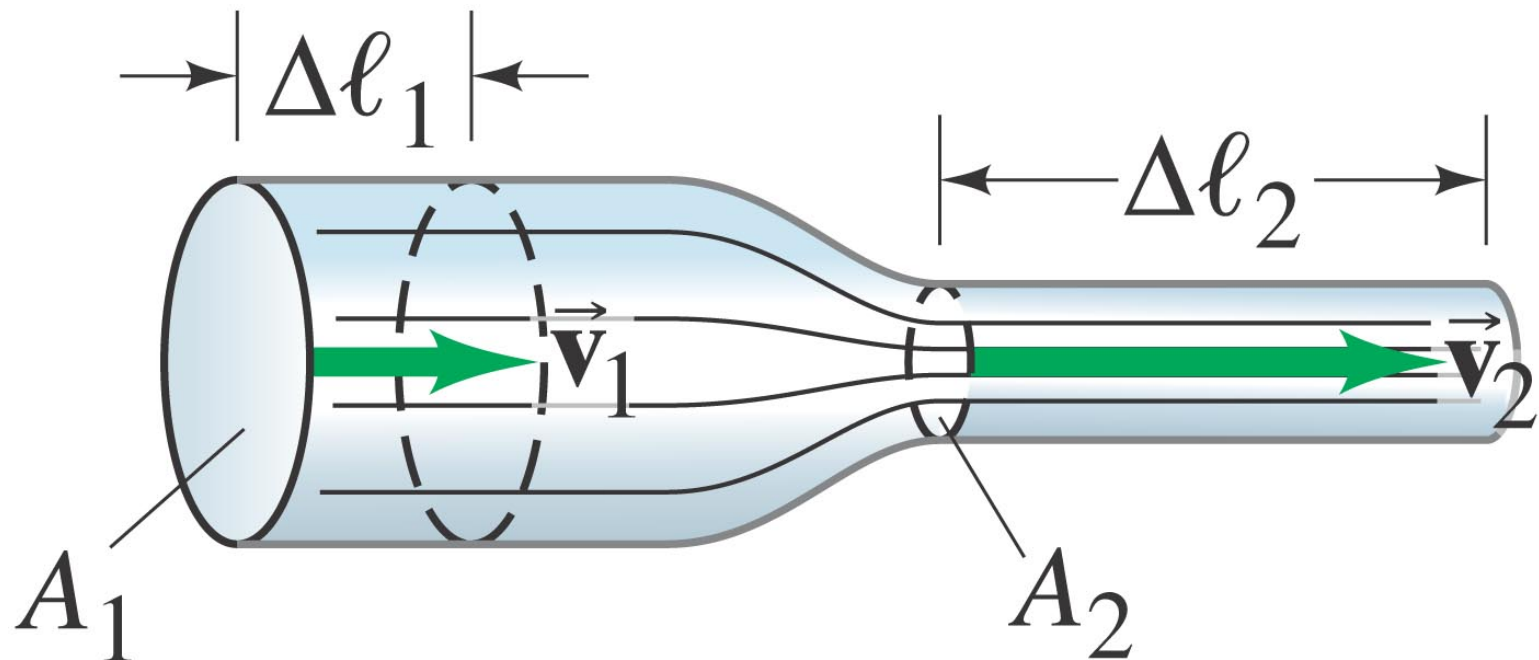
This gives us the **equation of continuity**:

Since
$$\frac{\Delta m_1}{\Delta t} = \frac{\Delta m_2}{\Delta t},$$

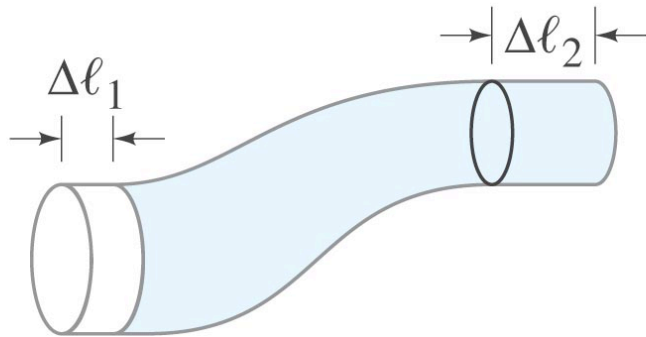
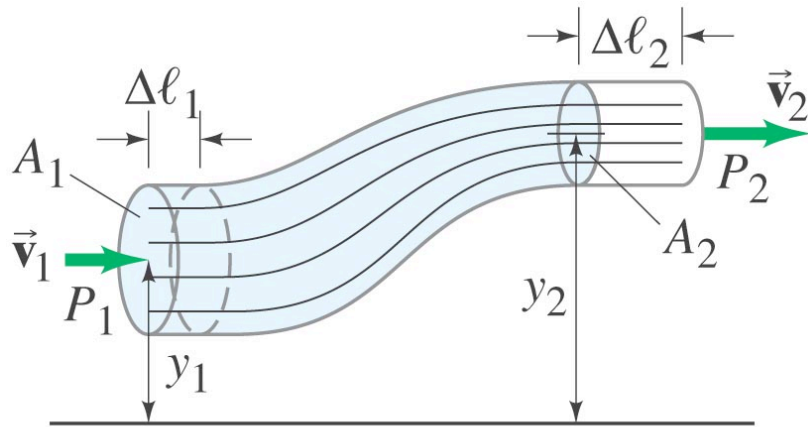
then
$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2.$$

13-8 Fluids in Motion; Flow Rate and the Equation of Continuity

If the density doesn't change—typical for liquids—this simplifies to $A_1 v_1 = A_2 v_2$. Where the pipe is **wider**, the flow is **slower**.



13-9 Bernoulli's Equation



Bernoulli's principle:

Where the velocity of a fluid is high, the pressure is low, and where the velocity is low, the pressure is high.

This makes sense, as a force is required to accelerate the fluid to a higher velocity.

13-9 Bernoulli's Equation

Consider the work it takes to move a small volume of fluid from one point to another while its flow is laminar. Work must be done to accelerate the fluid, and also to increase its height. Conservation of energy gives Bernoulli's equation:

$$\frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2 = P_1 - P_2 - \rho g y_2 + \rho g y_1,$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

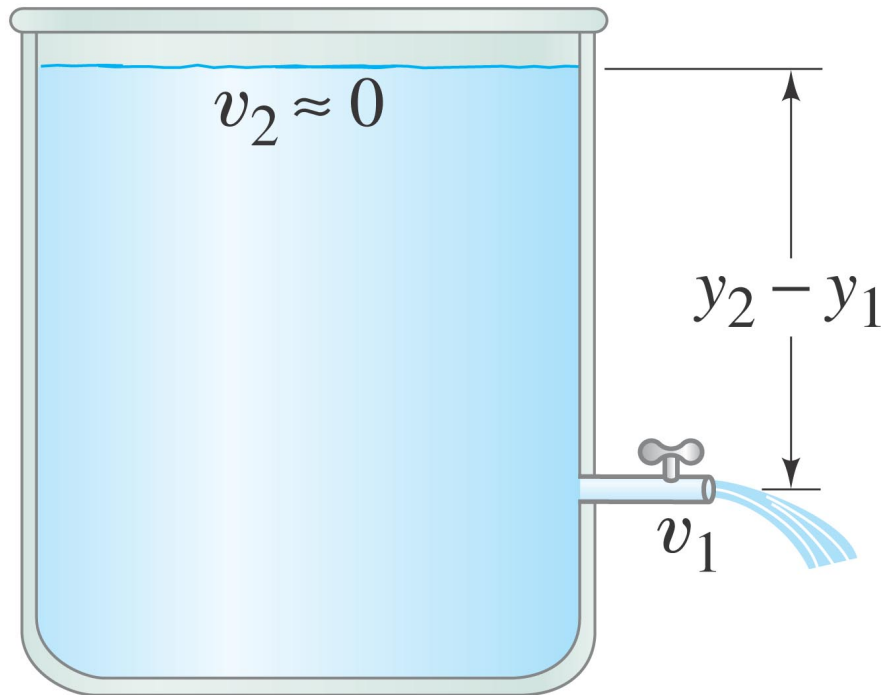
13-9 Bernoulli's Equation

Example 13-15: Flow and pressure in a hot-water heating system.

Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.5 m/s through a 4.0-cm -diameter pipe in the basement under a pressure of 3.0 atm , what will be the flow speed and pressure in a 2.6-cm -diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

13-10 Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA

Using Bernoulli's principle, we find that the speed of fluid coming from a **spigot on an open tank** is:



or

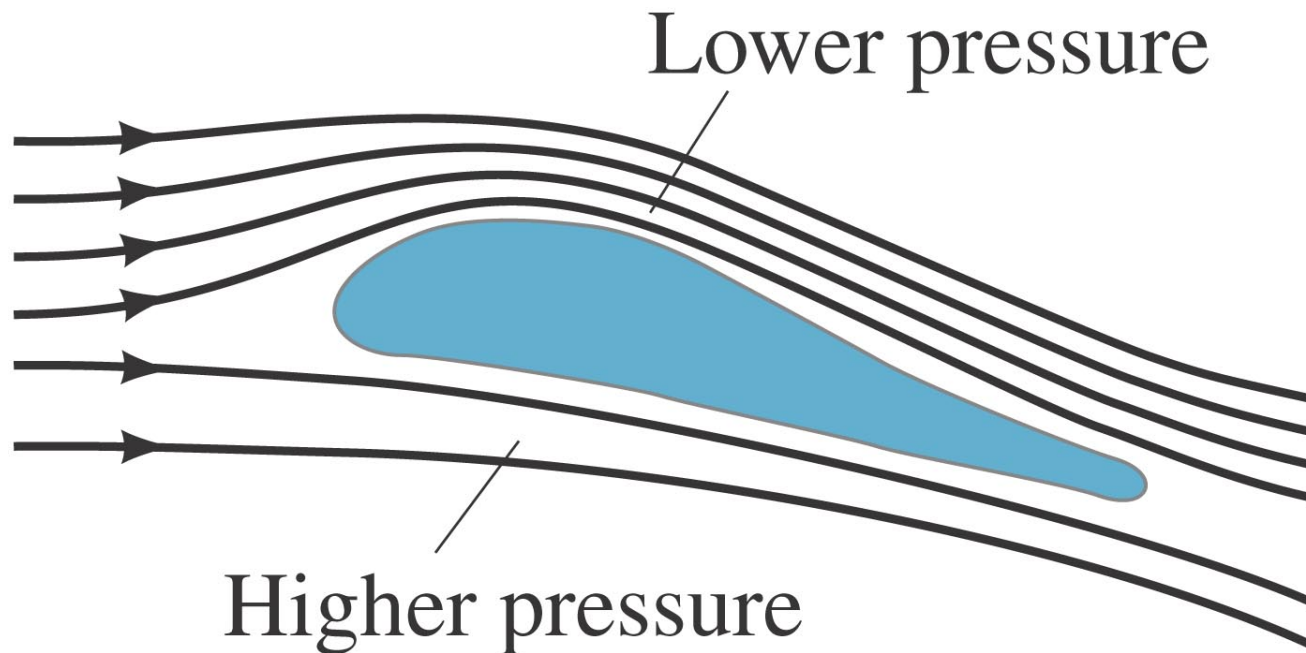
$$\frac{1}{2}\rho v_1^2 + \rho g y_1 = \rho g y_2$$

$$v_1 = \sqrt{2g(y_2 - y_1)}.$$

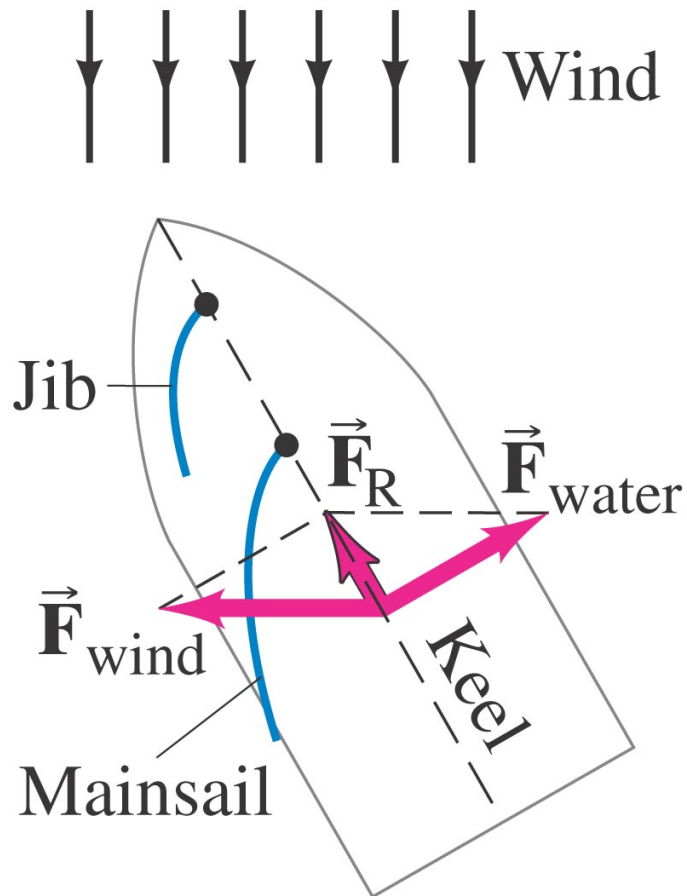
**This is called
Torricelli's theorem.**

13-10 Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA

Lift on an airplane wing is due to the different air speeds and pressures on the two surfaces of the wing.

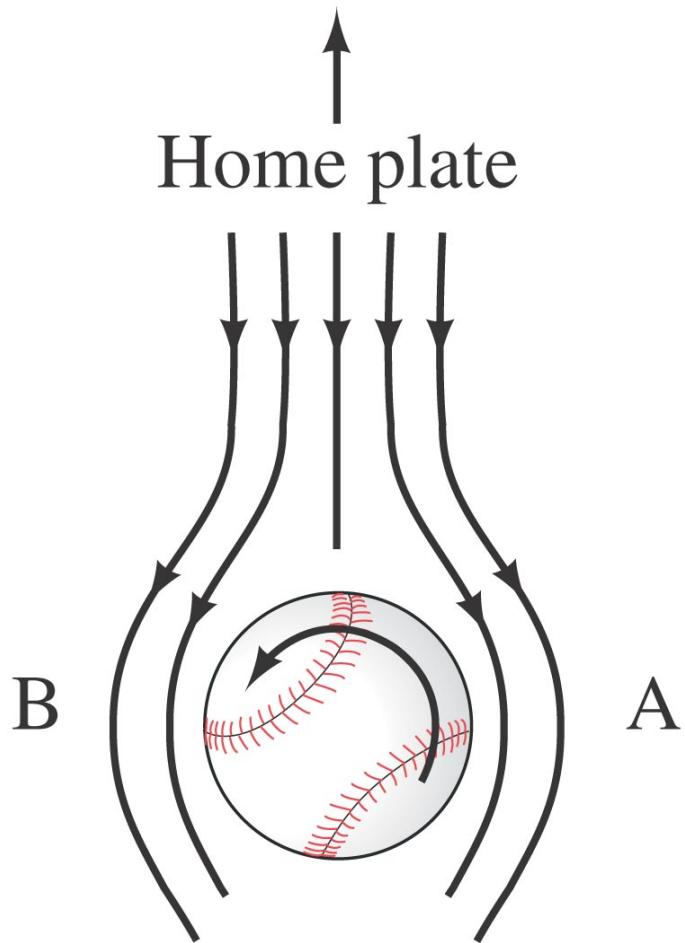


13-10 Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA



A sailboat can move against the wind, using the pressure differences on each side of the sail, and using the keel to keep from going sideways.

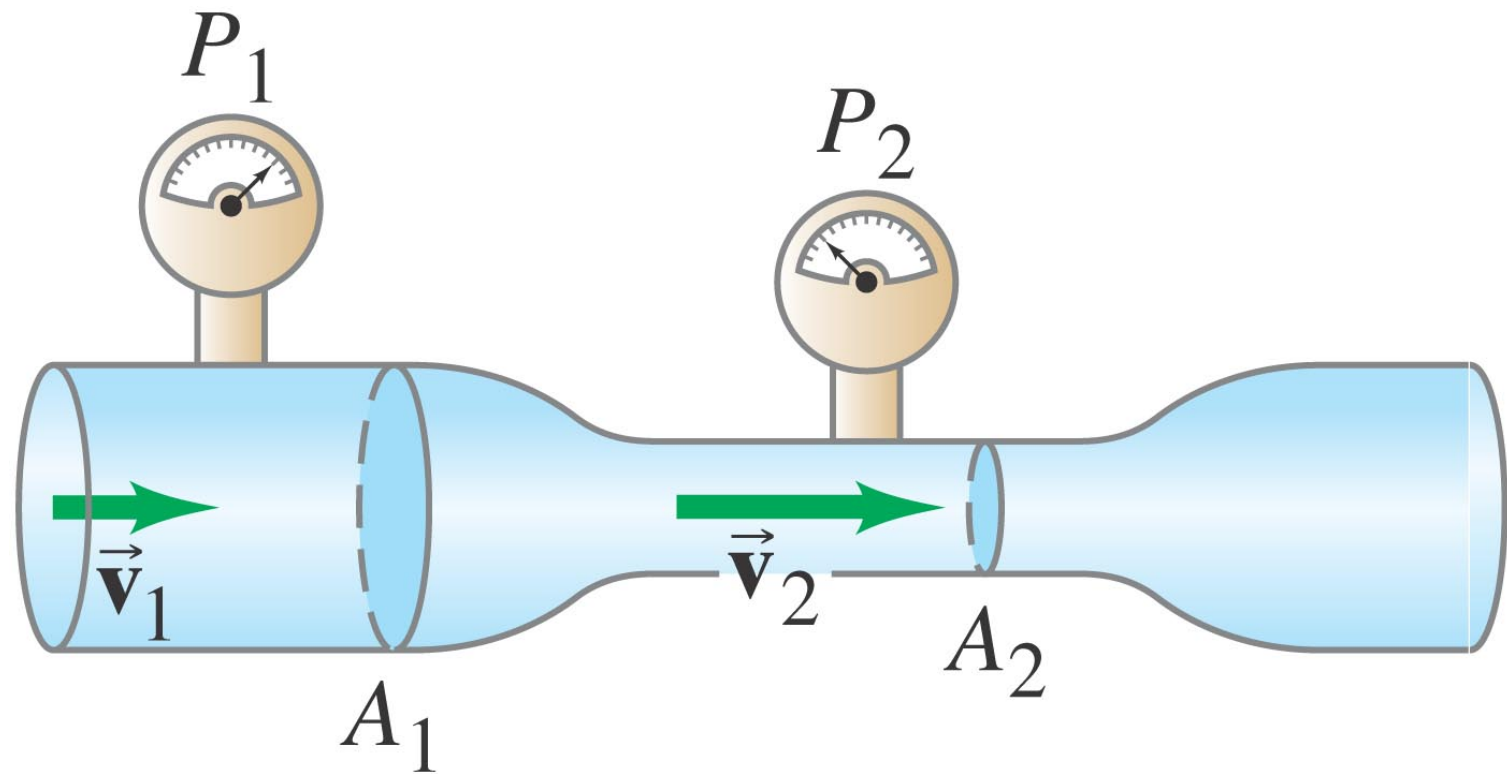
13-10 Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA



A ball's path will **curve** due to its **spin**, which results in the air speeds on the two sides of the ball not being equal; therefore there is a pressure difference.

13-10 Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA

A venturi meter can be used to measure fluid flow by measuring pressure differences.



Summary of Chapter 13

- Phases of matter: solid, liquid, gas
- Liquids and gases are called fluids.
- Density is mass per unit volume.
- Specific gravity is the ratio of the density of the material to that of water.
- Pressure is force per unit area.
- Pressure at a depth h is ρgh .
- External pressure applied to a confined fluid is transmitted throughout the fluid.

Summary of Chapter 13

- Atmospheric pressure is measured with a barometer.
- Gauge pressure is the total pressure minus the atmospheric pressure.
- An object submerged partly or wholly in a fluid is buoyed up by a force equal to the weight of the fluid it displaces.
- Fluid flow can be laminar or turbulent.
- The product of the cross-sectional area and the speed is constant for horizontal flow.

Summary of Chapter 13

- Where the velocity of a fluid is high, the pressure is low, and vice versa.
- Viscosity is an internal frictional force within fluids.
- Liquid surfaces hold together as if under tension.